

Buddy system wiring – power distribution and interconnect (nets by label)

Generated by tools/kicad/gen_system_wiring.py from design_params.yaml – regenerate, don't hand-edit.
Connectivity machine-checked by tools/kicad/check_system_wiring.py.

Companion sheet: electronics/KICAD/drive_mcu_wiring/ (pin-level drive harness). The two sheets share +12V_BUS, GND_PWR, +5V_RAIL, +3V3, GND and ESTOP_STATE – follow a rail across both by net name.

Protection chain: driver limit 8A/motor (firmware) -> BMS >= 50A -> fuse 60A -> E-stop contactor. ADR-0005.
The E-stop opens +12V_BUS itself; K1's AUX_NC contact only REPORTS the state to the MCU on PB12.
The LD19 is powered from the Jetson USB port through the D300 adapter, not from +5V_RAIL.
BNO086 sits on the Jetson 40-pin header (+3V3_JETSON), a different 3.3V rail from the Nucleo LDO's +3V3.
GND_PWR is the motor/battery return, starred at the pack; GND is the logic return. Single-point tie at the pack.

